

Name: <u>Solution</u>	ID:	Section:
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1.CO1	Divide $(3243)_5$ by $(12)_5$. Show all the working steps.	4
2.CO1	Add $(56)_{10}$ with $-(71)_{10}$ in 10 bits using 2's complement number system. Justify whether there is an overflow or not.	5
3.CO1	Suppose you are going on a picnic with $(1F)_{16}$ other people. Each person, including yourself, needs to contribute $(245)_6$ taka for the picnic expenses. After the picnic trip was over, you found out that $(33002)_4$ was spent on food and $(10101011)_2$ was spent on transportation. How much money was left after paying all the expenses? Show your answer in decimal.	6

$$\begin{array}{r}
 224 \\
 12 \overline{) 28243} \\
 \underline{-24} \\
 034 \\
 \underline{-24} \\
 103 \\
 \underline{-103} \\
 0
 \end{array}$$

$$\begin{aligned}
 12 \times 1 &= 12 \\
 12 \times 2 &= 24 \\
 12 \times 3 &= 36 \\
 12 \times 4 &= 48
 \end{aligned}$$

Quotient = $(224)_5$
Remainder = $(0)_5$

$$\begin{array}{r}
 2 \overline{) 56} \\
 2 \overline{) 28} \\
 2 \overline{) 14} \\
 2 \overline{) 7} \\
 2 \overline{) 3} \\
 2 \overline{) 1} \\
 0
 \end{array}$$

$$\begin{array}{r}
 2 \overline{) 71} \\
 2 \overline{) 35} \\
 2 \overline{) 17} \\
 2 \overline{) 8} \\
 2 \overline{) 4} \\
 2 \overline{) 2} \\
 2 \overline{) 1} \\
 0
 \end{array}$$

$(56)_{10} = (111000)_2$
 $(+56)_{10}$ in 10-bits = $(0000111000)_{2s}$

$(71)_{10} = (1000111)_2$
 $(+71) = 0001000111$
 $(-71) = 1110111000$

$$\begin{array}{r}
 0000\overset{1}{1}\overset{1}{1}000 \\
 + 1110\overset{1}{1}\overset{1}{1}001 \\
 \hline
 111110001
 \end{array}$$

$$\text{Result} = (111110001)_{25}$$

No overflow, since we are adding two different signed numbers.

$$3) (1F)_{16} = 1 \times 16^1 + 15 \times 16^0 = (31)_{10}$$

$$\text{Total people} = 31 + 1 = (32)_{10}$$

$$\text{Each person} = (245)_6 = 2 \times 6^2 + 4 \times 6^1 + 5 \times 6^0 = 101$$

contributes

$$\text{Total money} = 32 \times 101 = (3232)_{10}$$

$$\text{Food} = (33002)_4 = 3 \times 4^4 + 3 \times 4^3 + 2 \times 4^0 = (962)_{10}$$

$$\text{Transport} = (101010111)_2 = 1 \times 2^8 + 1 \times 2^6 + 1 \times 2^4 + 1 \times 2^2 + 1 \times 2^1 + 1 \times 2^0 = (343)_{10}$$

$$\begin{aligned} \text{Total money left} &= 3232 - 962 - 343 \\ &= \del{1927} (1927)_{10} \end{aligned}$$